ARTIFICIAL INTELLIGENCE (AI)

OVERVIEW

Applying leadership and 21st century skills, participants demonstrate their knowledge of Artificial Intelligence (AI) concepts, models, and best practices by completing a test in order to qualify for the semifinals. Semifinalist teams demonstrate 21st century skills through participating in solving onsite problems using a GenAI tool. Completed solutions are objectively measured to determine the best solution for the stated problem.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

TIME LIMITS

PRELIMINARY ROUND

A. One (1)-hour test is administered to all members at the same time.

SEMIFINAL ROUND

A. Seventy-five (75) minutes for the planning and solutioning are permitted.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

PRELIMINARY ROUND

A. Participants report to the event area at the time and place stated in the conference program for the test.

B. Participants follow the specific regulations and adhere to the directions provided on-site by the event coordinator.

C. All participants take the test at the same time and take the test individually.

D. The top twelve (12) scoring individuals qualify as semifinalists.

E. A semifinalist list (in random order) is posted.

SEMIFINAL ROUND

A. Participants bring their own computer systems to the event area at the time and place stated in the conference program.

B. Participants are given a problem, evaluation criteria, materials, and allotted seventy-five (75) minutes for the design and development of the solution.

C. Solutions are presented to the judges as soon as possible upon completion or when time expires – whichever comes first.

D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Participants take the test individually and without any external resources or help.

B. Tests may be administered online or via a scan-type answer sheet. Please review the *Competition Updates* page on the TSA website.

C. Participants are responsible for bringing:

1. One (1) laptop or tablet (personal or school-issued) capable of networking via Wi-Fi and able to run solely on battery power for up to two (2) consecutive hours. Chrome is the preferred browser.
2. Optional: One (1) mouse
3. Two (2) sharpened No.2 pencils

D. Scan-type forms are furnished by the event coordinator, if applicable.

E. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.

F. Participant identification numbers must be entered on the scan form in the space indicated.

G. Failure to follow instructions will result in the score sheet not being scored.

H. Participants must stop work immediately when time is called.

I. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.

J. All tests must be turned in before leaving the test area.

SEMIFINAL ROUND

A. All work must be completed in the event area during the time specified for the event.

B. Each participant must bring:

1. one (1) laptop capable of networking via Wi-Fi with Microsoft Edge, Google Chrome, or Mozilla Firefox installed. The laptop must be fully charged with enough battery for the duration of the semifinal round.
2. pencils, paper, and an external mouse are recommended but not required for each participant.
3. portable power bank (if desired).

C. External keyboard and monitors are not permitted.

D. Printed reference materials are not allowed.

E. Access to Internet will not be provided. Teams may provide and use a hotspot during the semifinal round. If using a student provided hotspot, National TSA does not guarantee reliable connectivity during the event.

F. Participants do NOT have access to electrical power/outlets during the event.

G. Participants must have all software necessary for the competition downloaded and accessible on their laptop.

H. Each participant will create an individual PDF for each solution that specifies the Generative AI tools used for each problem.

I. All solutions must be tested, demonstrated, and presented by the participant in front of the judges exclusively through electronic submission and evaluation.

EVALUATION

PRELIMINARY ROUND

A. Test

SEMIFINAL ROUND

A. On-site solution

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

Depending upon the subject of the problem, this event may align with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Information & support services
- Web & digital communications
- Cybersecurity Engineer

Participant/Team ID# _____

ARTIFICIAL INTELLIGENCE (AI)

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Computer hardware is present (semifinals only)
☐ ENTRY NOT EVALUATED

TEST SCORE (50 points)	
Average of the two (2) team members' test scores.	**TEST SCORE SUBTOTAL** (50 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY ROUND SUBTOTAL (50 points)

TESTING OF SOLUTION (80 points)					
Evaluation: A finite unit of measure, such as time and/or number correct, etc., is used to determine ranking.					
1st: 80 Points	**2nd:** 75 Points	**3rd:** 70 Points	**4th**: 65 Points	**5th:** 60 Points	**6th:** 55 Points
7th: 50 Points	**8th:** 45 Points	**9th:** 40 Points	**10th:** 35 Points	**11th:** 30 Points	**12th:** 25 Points
					TESTING OF SOLUTION SUBTOTAL (80 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SOLUTION SUBTOTAL (80 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	**TOTAL** (130 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ARTIFICIAL INTELLIGENCE (AI) EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator

B. Judges, two (2) or more

C. Assistants for set-up, monitoring, and clean-up of on-site activity, two (2) or more

D. Proctors for exam, three (3)

MATERIALS

A. Coordinator's packet, containing:

   1. Event guidelines, one (1) copy for the coordinator and for each judge
   2. TSA Event Coordinator Report

B. If hard copies of the test are being used, have one (1) for each participant (these tests must be returned immediately following the event).

C. Scantron answer sheets; one (1) for each participant.

D. Tables and chairs for participants

E. Tables and chairs for judges, to be used for information distribution and evaluation

F. A copy of a well-written, technologically appropriate problem for each participating team and each judge that can be measured

G. Adequate conditions, tools, materials, monitoring, and testing devices for the problem

   1. Stopwatch for timekeeper
   2. Tables and chairs or tablet armchairs to accommodate all participants
   3. Scantron instruction forms

H. Coordinators are responsbile for creating the test to be administered at the National TSA Conference; copies are provided by the national TSA office.

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.

B. Report to the CRC room and check the contents of the coordinator's packet.

C. Review the event guidelines and check to see that enough personnel have been scheduled.

D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

E. At least one (1) hour before the event is to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details pertaining to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.

F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

A. Begin the event at the scheduled time by closing the doors and checking the entry list.

B. All participants and event judges should be in the room at this time.

C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.

D. In order to compete, participants must be on the entry list or must have approval of the CRC.

E. Distribute the scantron forms to the participants, if applicable.

   1. Direct participants to fill in their participant identification number and test code letter in the appropriate spaces.
   2. Provide an opportunity for any questions about the scan form.

F. Ensure the following testing procedure is applied with the help of the proctors.
   1. If the test is administered as hard copies, instruct the participants to keep the tests face down until they are directed to turn them over and begin.
   2. If exams are administered electronically, instruct participants not to begin until the scheduled time.

G. Acting as the timer and with proctors positioned around the event room, direct the participants to turn their test over, place their code number and the code letter found on the test on the scan form, and begin.

H. Exactly one (1) hour from the time that the participants begin the test, call time.
   1. Direct students to check out with a test proctor once they are finished with their test.
   2. Proctors should collect all tests and then students should immediately leave the testing room.
   3. If a line forms, students must remain completely silent. Any talking will result in a zero score for test of the person(s) talking.

I. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and manager to determine either:
   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

J. Post the list of the twelve (12) semifinalist teams.

SEMIFINAL ROUND

A. Begin the event at the scheduled time by closing the doors and checking the entry list.

B. All participants and judges should be in the room at this time.

C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond participant control.

D. In order to compete, participants must be on the entry list or must have approval of the CRC.

E. Check in the semifinalist teams and equipment. Participants must bring:
   1. One (1) laptop
   2. Teams may also bring pencils and paper.
   3. Portable power bank (if desired)

F. Teams will not have access to electrical power/outlets during the event.

G. Access to Internet will not be provided. Teams may proivde and use a hotspot during the semifinal round. If using a student provided hotspot, National TSA does not guarantee reliable connectivity during the event.

H. Students must have all software development tools needed for the competition downloaded and accessible on their computers.

I. Once teams are seated and general announcements have been given, distribute and review the problem and start the time.

J. Judges will evaluate the submissions upon completion.

K. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

L. If necessary, manage security and the removal of materials from the area.